

FLAVIO MARTINELLI

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EDUCATION

last updated July 30, 2026

2021-2026	EPFL , Ph.D. in Computational Neuroscience, Lausanne (CH) <i>Advisors: Prof. Wulfram Gerstner & Ph.D. Johanni Brea</i>	
2025	Harvard , Visiting Doctoral Student, Boston (US) <i>Kempner Institute, Harvard Medical School; advisor: Prof. Kanaka Rajan</i>	
2020	EPFL , M.Sc. in Life Sciences and Technologies, Lausanne (CH)	GPA: 5.55/6.00
2017	Politecnico di Milano , B.Eng. in Biomedical Engineering, Milano (IT)	GPA: 109/110

WORK EXPERIENCE

2020	CHUV , Research Assistant at university hospital (LREN lab), Lausanne (CH) <i>Reinforcement learning modelling of fMRI data</i>
2019	Logitech , R&D Intern, Lausanne (CH) <i>Neuromorphic computing for speech detection</i>

SELECTED PUBLICATIONS

2026	Preprint : “The Puzzling Success of Overparameterization: Lottery Tickets or Escape Dimensions?” Martinelli, F., Brea J. & Gerstner W.
2025	NeurIPS 2025 : “Flat Channels to Infinity in Neural Loss Landscapes” Martinelli, F.*, Van Meegen A.*, Şimşek B., Gerstner W. & Brea J.
2025	Nature Machine Intelligence : “Actor-Critic Networks with Analogue Memristors Mimicking Reward-Based Learning” Portner K.*, Zellweger T.*, Martinelli, F., ..., Offrein B., Gerstner W., Luisier M. & Emboras A.
2024	ICML 2024 : “Expand-and-Cluster: Parameter Recovery of Neural Networks” Martinelli, F., Şimşek B., Gerstner W.* & Brea J.*

* equal contribution

HONORS AND AWARDS

2023	Best Presentation Award , NeuroLeman Annual Meeting, Villars (CH)
2020	Mention of Excellence for final GPA, EPFL, Lausanne (CH)
2014-2017	High Merits tuition exemptions, Politecnico di Milano (IT)
2013	First Place at National Electronics Competition , Bergamo (IT)

SKILLS

CODING	Python, Julia, MATLAB, C/C++
TOOLS	PyTorch, Git, Shell, Kubernetes
SOFT	Teaching, Project design, Supervision

LANGUAGES

ITALIAN	Native
ENGLISH	Professional
FRENCH	Intermediate

INVITED TALKS

- 2025 *“Flat Channels to Infinity in Neural Loss Landscapes”*
 Graz University of Technology, Graz (AT), ELLIS Math&Efficiency reading group
 “Flat Channels to Infinity in Neural Loss Landscapes”
 EPFL, Lausanne (CH), ELLIS pre-NeurIPS event
 “Flat Channels to Infinity in Neural Loss Landscapes”
 Online, Ploutos platform
 “Reverse Engineering Neural Circuits: Insights from Loss Landscape Geometry”
 Harvard University, Boston (USA), Kanaka Rajan’s group
- 2024 *“Expand-and-Cluster: Parameter Recovery of Neural Networks”*
 Graz University of Technology, Graz (AT), EfficientML reading group
 “From Loss Landscape Geometry to Weight Recovery in Neural Networks”
 Institute of Science and Technology, Vienna (AT), Tim Vogels’s group
 “Expand-and-Cluster: Parameter Recovery of Neural Networks”
 Max Planck Institute for Intelligent Systems, Tübingen (DE), Jakob Macke’s group
 “Expand-and-Cluster: Parameter Recovery of Neural Networks”
 Crans-Montana (CH), Swiss Computational Neuroscience Retreat 2024
- 2023 *“Expand-and-Cluster: Parameter Recovery of Neural Networks”*
 Villars (CH), Annual Meeting of the NeuroLeman Network 2023
 “Expand-and-Cluster: Parameter Recovery of Neural Networks”
 ETH, Zurich (CH), Angelika Steger and Joao Sacramento’s groups

COURSES AND WORKSHOPS

- 2026 **Organizer NeurIPS AXIOM workshop**, Paris (FR)
- 2025 **MIT Brains, Minds and Machines** summer course, Woods Hole, MA (USA)

CONFERENCE ACTIVITIES

- 2025 **UniREPS**: poster, San Diego (USA)
 NeurIPS: poster, San Diego (USA)
 Bernstein Conference: poster, Frankfurt (DE)
 Frontiers in NeuroAI: poster, Boston (USA)
 Spring into Science (Harvard Kempner Institute): poster, Boston (USA)
- 2024 **ICML**: poster, Vienna (AT)
 Youth in High-Dimensions: poster, Trieste (IT)
- 2023 **NeuroLeman Meeting**: talk and poster, Villars (CH)
 Bernstein Conference: poster, Berlin (DE)
- 2022 **Bernstein Conference**: poster, Berlin (DE)

TEACHING EXPERIENCE (TA)

- 2023 **Computational Neuroscience: Neuronal Dynamics**, EPFL master course
- 2022 **Biological Modelling of Neural Networks**, EPFL master course
- 2021-2023 **Introduction to Machine Learning for Bioengineers**, EPFL bachelor course
- 2021- **Supervised 5 Master Students**:
 R. Palazzo, A. Salvatore, A. Beiser, M. Brodeur, R. Jabyiev

- 2025 **NeurIPS 2025 (spotlight):** “*Measuring and Controlling Solution Degeneracy across Task-Trained Recurrent Neural Networks*”
Huang A., Singh S.H., Martinelli, F. & Rajan K.
NeurIPS UniReps workshop: “*Data Augmentation Techniques to Reverse-Engineer Neural Network Weights from Input-Output Queries*”
Beiser A., Martinelli, F., Gerstner W. & Brea J.
- 2023 **ArXiv:** “*MLPGradientFlow: going with the flow of multilayer perceptrons (and finding minima fast and accurately)*”
Brea J., Martinelli, F., Şimşek B. & Gerstner W.
- 2020 **ICAASP 2020:** “*A Bin Encoding Training of a Spiking Neural Network Based Voice Activity Detection*”
Dellaferrera G., Martinelli, F. & Cernak M.
ICAASP 2020: “*Spiking neural networks trained with backpropagation for low power neuromorphic implementation of voice activity detection*”
Martinelli, F., Dellaferrera G., Mainar P. & Cernak M.